

git log/history

References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

Git History - Directory Commit Statistics

Treemap Layout Functions and Constants

Visualization Data Preparation Functions

File Data Preparation Functions

File Data Preparation

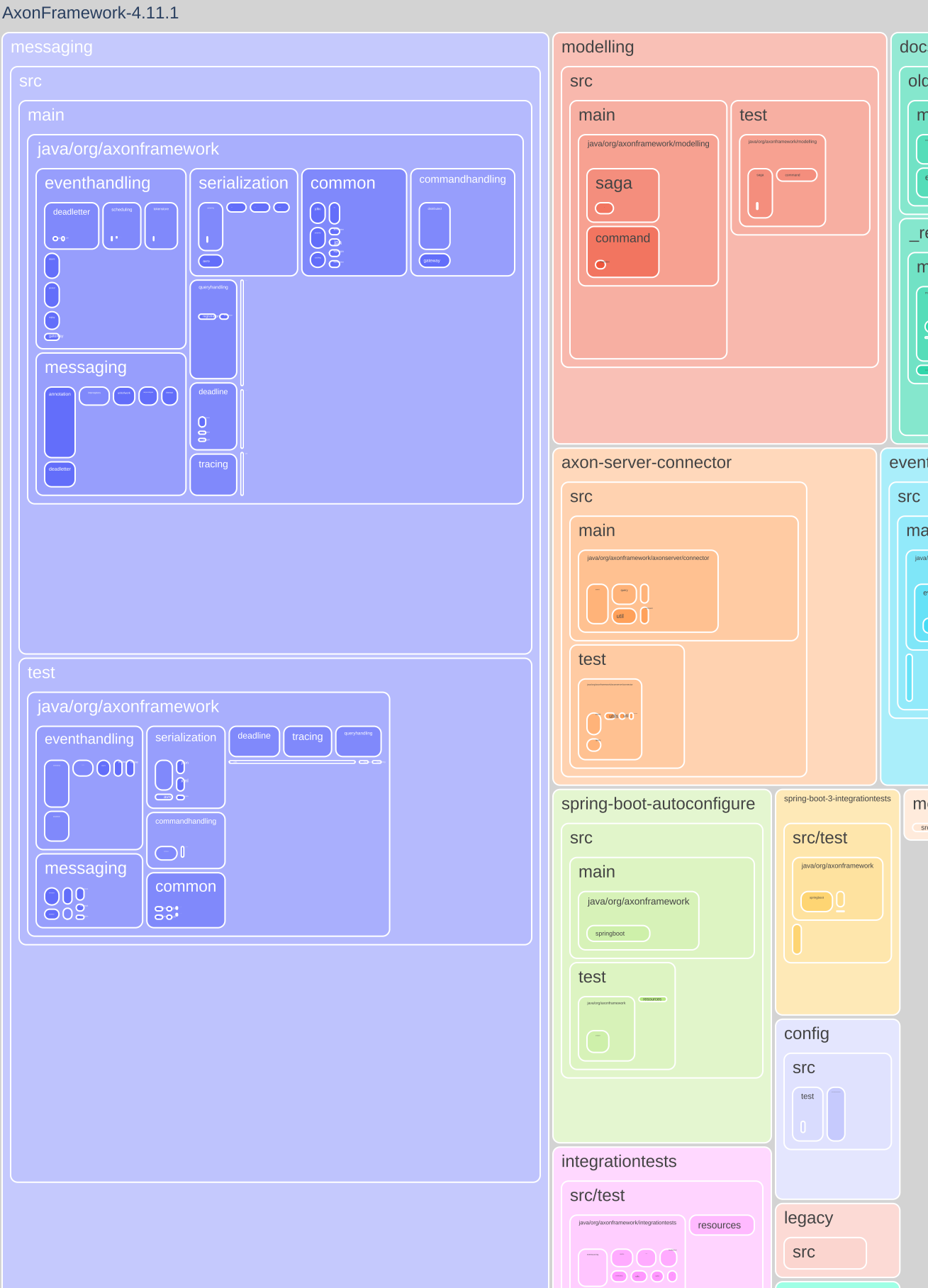
Data Preview

	fileCount	authorCount	commitCount	daysSinceLastCommit	daysSinceLastCreation	daysSinceLastModification
count	563.000000	563.000000	563.000000	563.000000	563.000000	563.000000
mean	24.492007	19.502664	1596.113677	204.362345	1062.840142	298.296625
std	117.928146	21.209040	7954.335528	127.121149	968.683356	367.528519
min	1.000000	2.000000	3.000000	57.000000	64.000000	57.000000
25%	1.000000	7.000000	50.500000	121.000000	304.000000	121.000000
50%	4.000000	12.000000	141.000000	263.000000	823.000000	263.000000
75%	10.000000	25.000000	595.500000	263.000000	1877.000000	263.000000
max	2250.000000	212.000000	154891.000000	2413.000000	5541.000000	2892.000000

	directoryPath	directoryParentPat
0	AxonFramework-4.11.1/assembly	AxonFramework-4.11.
1	AxonFramework-4.11.1/axon-server-connector/src/main/java/org/axonframework/axonserver/connector/heartbeat/source	AxonFramework-4.11.1/axon-serve
2	AxonFramework-4.11.1/axon-server-connector/src/main/resources/META-INF/services	AxonFramework-4.11.1/axon-server-connector/src/main/resource
3	AxonFramework-4.11.1/axon-server-connector/src/test/java/org/axonframework/axonserver/connector/event/util	AxonFramework-4.11.1/axon-serve
4	AxonFramework-4.11.1/config/src/test/resources/META-INF	AxonFramework-4.11.1/config/src/test/resource
5	AxonFramework-4.11.1/coverage-report	AxonFramework-4.11.
6	AxonFramework-4.11.1/docs/_playbook/localLinks	AxonFramework-4.11.1/docs/_playbo
7	AxonFramework-4.11.1/docs/_reference/modules/ROOT/attachments	AxonFramework-4.11.1/docs/_reference/modules/ROO
8	AxonFramework-4.11.1/docs/_reference/modules/ROOT/partials	AxonFramework-4.11.1/docs/_reference/modules/ROO
9	AxonFramework-4.11.1/docs/_reference/modules/axon-server-connector/pages	AxonFramework-4.11.1/docs/_reference/modules/axon-server-connect
10	AxonFramework-4.11.1/docs/_reference/modules/axon-server-connector/partials	AxonFramework-4.11.1/docs/_reference/modules/axon-server-connect
11	AxonFramework-4.11.1/docs/_reference/modules/configuration/pages	AxonFramework-4.11.1/docs/_reference/modules/configuratio
12	AxonFramework-4.11.1/docs/_reference/modules/configuration/partials	AxonFramework-4.11.1/docs/_reference/modules/configuratio
13	AxonFramework-4.11.1/docs/_reference/modules/disruptor/pages	AxonFramework-4.11.1/docs/_reference/modules/disruptor
14	AxonFramework-4.11.1/docs/_reference/modules/disruptor/partials	AxonFramework-4.11.1/docs/_reference/modules/disruptor
15	AxonFramework-4.11.1/docs/_reference/modules/eventsourcing/pages	AxonFramework-4.11.1/docs/_reference/modules/eventssourcin
16	AxonFramework-4.11.1/docs/_reference/modules/eventsourcing/partials	AxonFramework-4.11.1/docs/_reference/modules/eventssourcin
17	AxonFramework-4.11.1/docs/_reference/modules/legacy/pages	AxonFramework-4.11.1/docs/_reference/modules/legac
18	AxonFramework-4.11.1/docs/_reference/modules/legacy/partials	AxonFramework-4.11.1/docs/_reference/modules/legac
19	AxonFramework-4.11.1/docs/_reference/modules/metrics/pages	AxonFramework-4.11.1/docs/_reference/modules/metric
20	AxonFramework-4.11.1/docs/_reference/modules/metrics/partials	AxonFramework-4.11.1/docs/_reference/modules/metric
21	AxonFramework-4.11.1/docs/_reference/modules/micrometer/pages	AxonFramework-4.11.1/docs/_reference/modules/micromete
22	AxonFramework-4.11.1/docs/_reference/modules/micrometer/partials	AxonFramework-4.11.1/docs/_reference/modules/micromete
23	AxonFramework-4.11.1/docs/_reference/modules/modeling/partials	AxonFramework-4.11.1/docs/_reference/modules/modelin
24	AxonFramework-4.11.1/docs/_reference/modules/spring/pages	AxonFramework-4.11.1/docs/_reference/modules/sprin
25	AxonFramework-4.11.1/docs/_reference/modules/spring/partials	AxonFramework-4.11.1/docs/_reference/modules/sprin
26	AxonFramework-4.11.1/docs/_reference/modules/springboot-starter/pages	AxonFramework-4.11.1/docs/_reference/modules/springboot-start
27	AxonFramework-4.11.1/docs/_reference/modules/springboot-starter/partials	AxonFramework-4.11.1/docs/_reference/modules/springboot-start
28	AxonFramework-4.11.1/docs/_reference/modules/test/pages	AxonFramework-4.11.1/docs/_reference/modules/te
29	AxonFramework-4.11.1/docs/_reference/modules/test/partials	AxonFramework-4.11.1/docs/_reference/modules/te

Number of files per directory

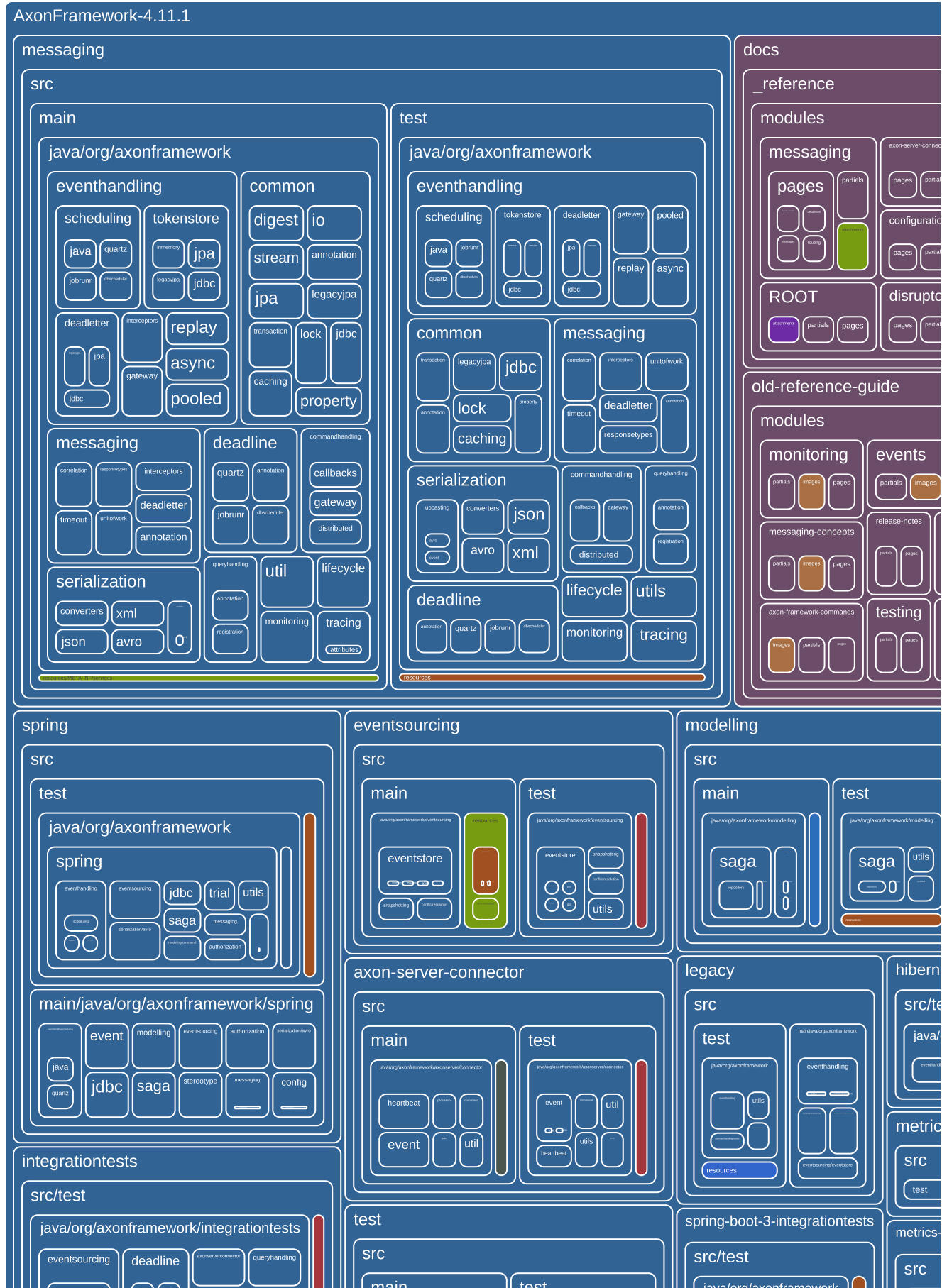
Directories and their file count

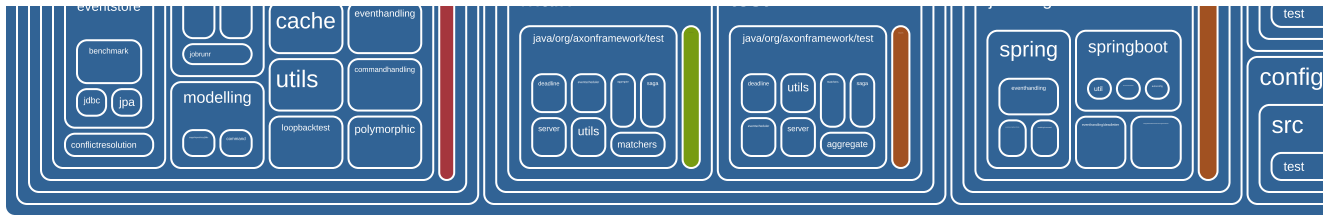




Most frequent file extension per directory

Most frequent file extension per directory

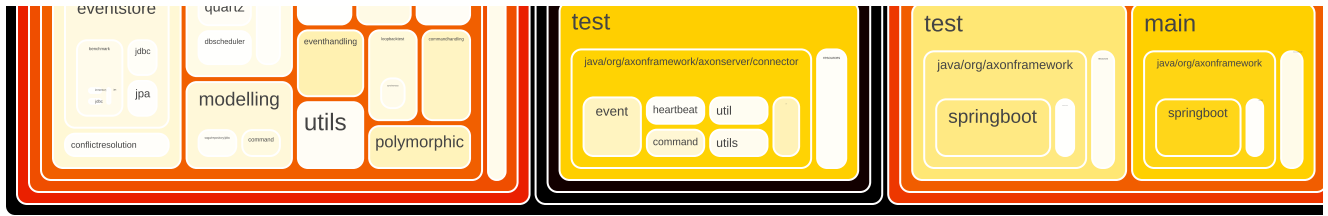




Number of commits per directory

Number of git commits

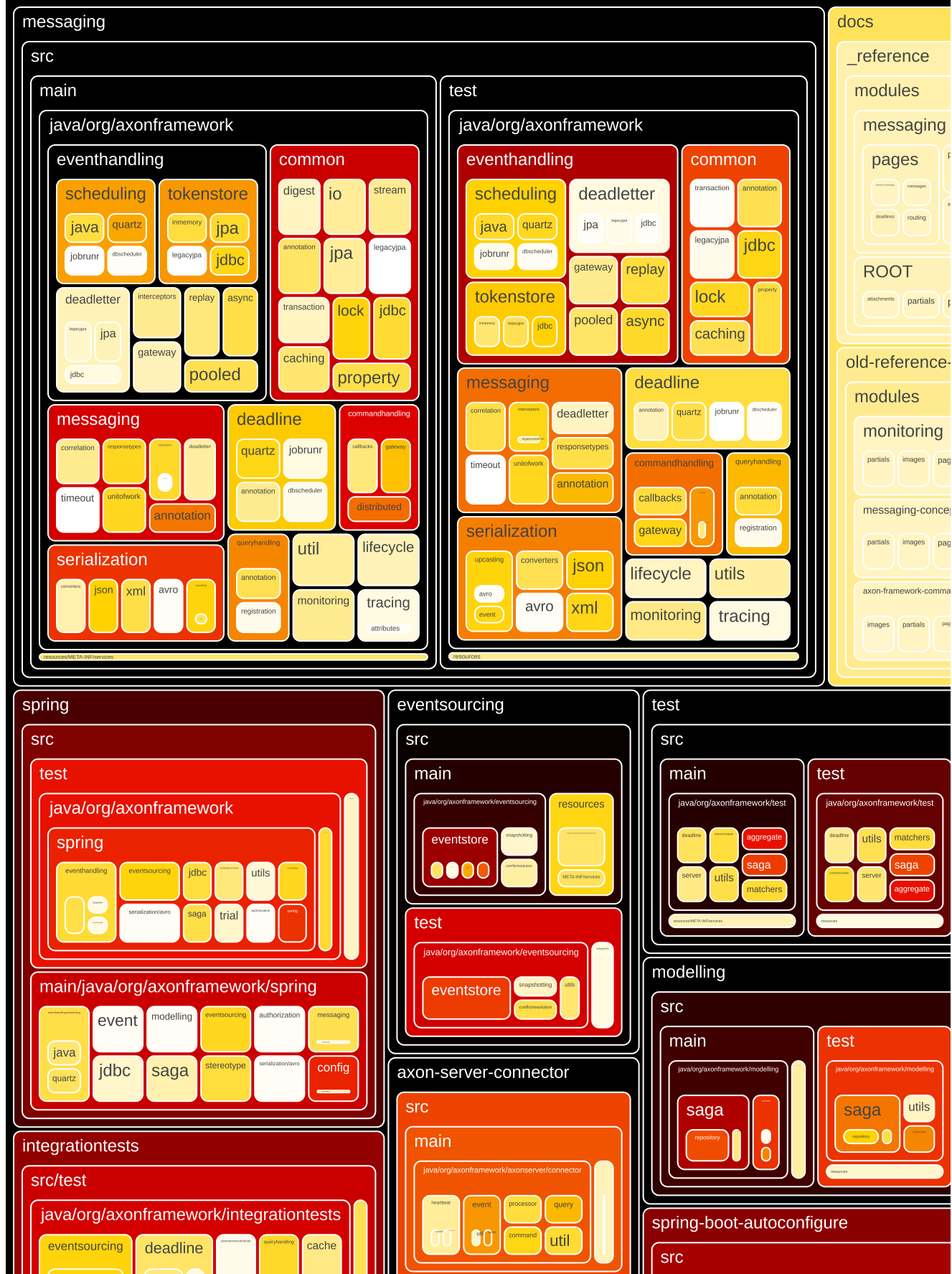


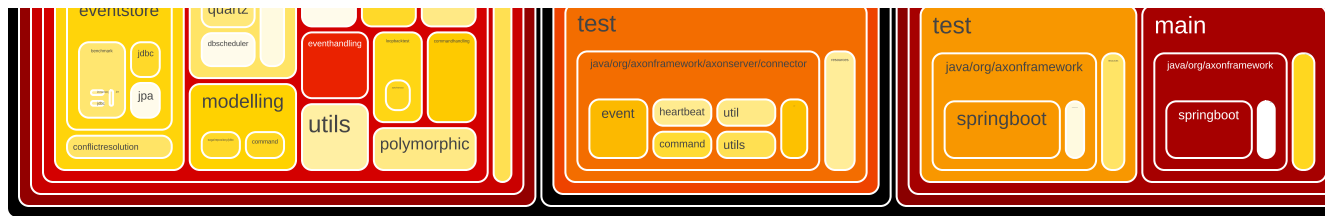


Number of distinct authors per directory

Number of distinct commit authors

AxonFramework-4.11.1





Directories with very few different authors

Number of distinct commit authors (red/black = only one or very few authors)

AxonFramework-4.11.1

messaging

src

main

java/org/axonframework

eventhandling

scheduling

tokenstore

common

digest

io

stream

java

quartz

inmemory

jpa

jobrunr

dscheduler

legacyjpa

jdbc

annotation

jpa

legacyjpa

deadletter

interceptors

replay

async

transaction

lock

jdbc

legacyjpa

jpa

gateway

pooled

caching

property

messaging

correlation

responsetypes

deadletter

timeout

unitofwork

annotation

serialization

converters

json

xml

avro

annotation

registration

monitoring

tracing

attributes

resources/META-INF/services

test

java/org/axonframework

eventhandling

scheduling

java

quartz

jobrunr

dscheduler

deadletter

gateway

pooled

tokenstore

inmemory

legacyjpa

jdbc

replay

async

messaging

correlation

interceptors

unitofwork

timeout

deadletter

responsetypes

annotation

serialization

upcasting

converters

json

avro

event

avro

xml

resources

common

transaction

annotation

legacyjpa

jdbc

lock

property

caching

deadline

annotation

quartz

commandhandling

queryhandling

callbacks

annotation

gateway

registration

lifecycle

utils

monitoring

tracing

docs

_reference

modules

messaging

pages

partials

attachments

ROOT

attachments

partials

pages

old-reference-guide

modules

monitoring

partials

images

pages

messaging-concepts

partials

images

pages

axon-framework-commands

images

partials

pages

spring

src

test

java/org/axonframework

spring

eventhandling

eventsourcing

jdbc

utils

event

modelling

saga

trial

authorization

main/java/org/axonframework/spring

event

modelling

eventsourcing

authorization

messaging

java

jdbc

saga

stereotype

serialization

config

integrationtests

src/test

java/org/axonframework/integrationtests

eventsourcing

deadline

queryhandling

cache

eventsourcing

src

main

java/org/axonframework/eventsourcing

eventstore

snapshotting

configuration

test

java/org/axonframework/eventsourcing

eventstore

snapshotting

utils

axon-server-connector

src

main

java/org/axonframework/axonserver/connector

heartbeat

event

processor

query

util

command

util

test

src

main

java/org/axonframework/test

annotation

server

utils

server

saga

test

java/org/axonframework/test

annotation

utils

server

annotation

saga

legacy

src

test

java/org/axonframework

eventhandling

commandhandling

spring-boot-3-integrationtests

src/test

modellir

src

main

java/org

saga

test

java/org

saga

hibernate-4

src/test

java/org

eventhandling

resources

eventstore
benchmark
jdbc
conflictresolution
quartz
eventhandling
modelling
utils
polymorphic

test

java/org/axonframework/axonserver/connector

event heartbeat util
command utils

src/test

java/org/axonframework

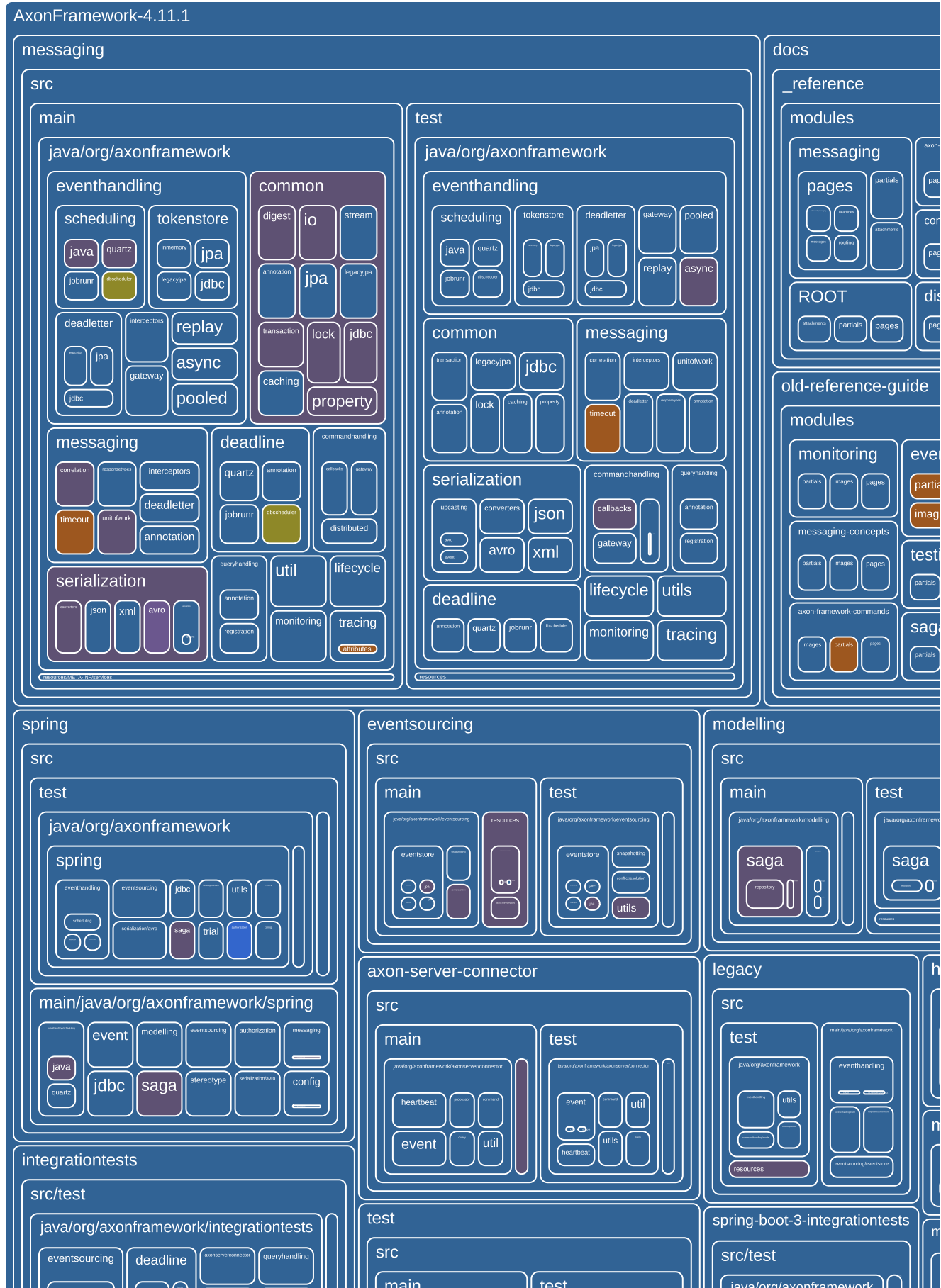


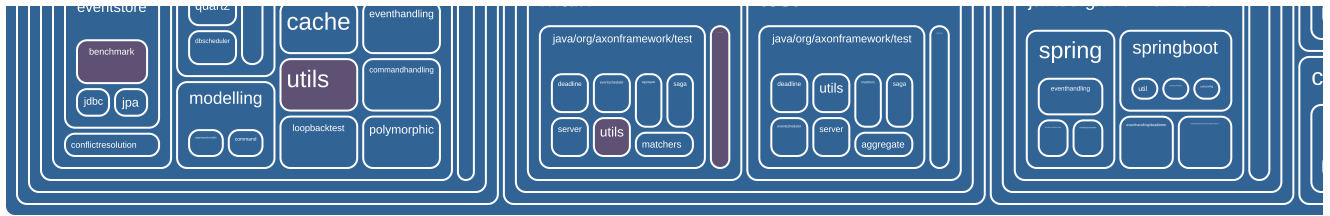
springboot



Main author per directory

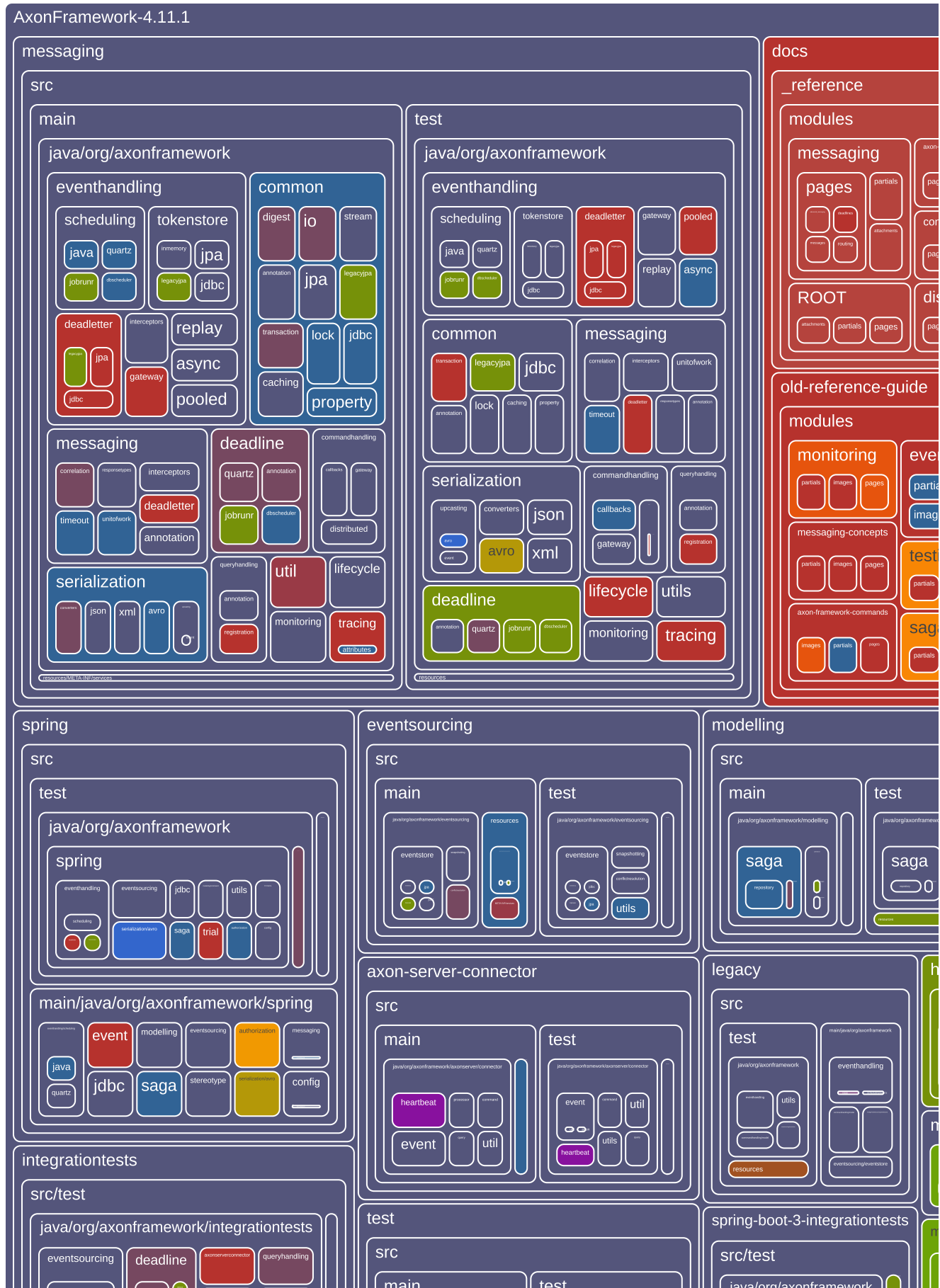
Main authors with highest number of commits





Second author per directory

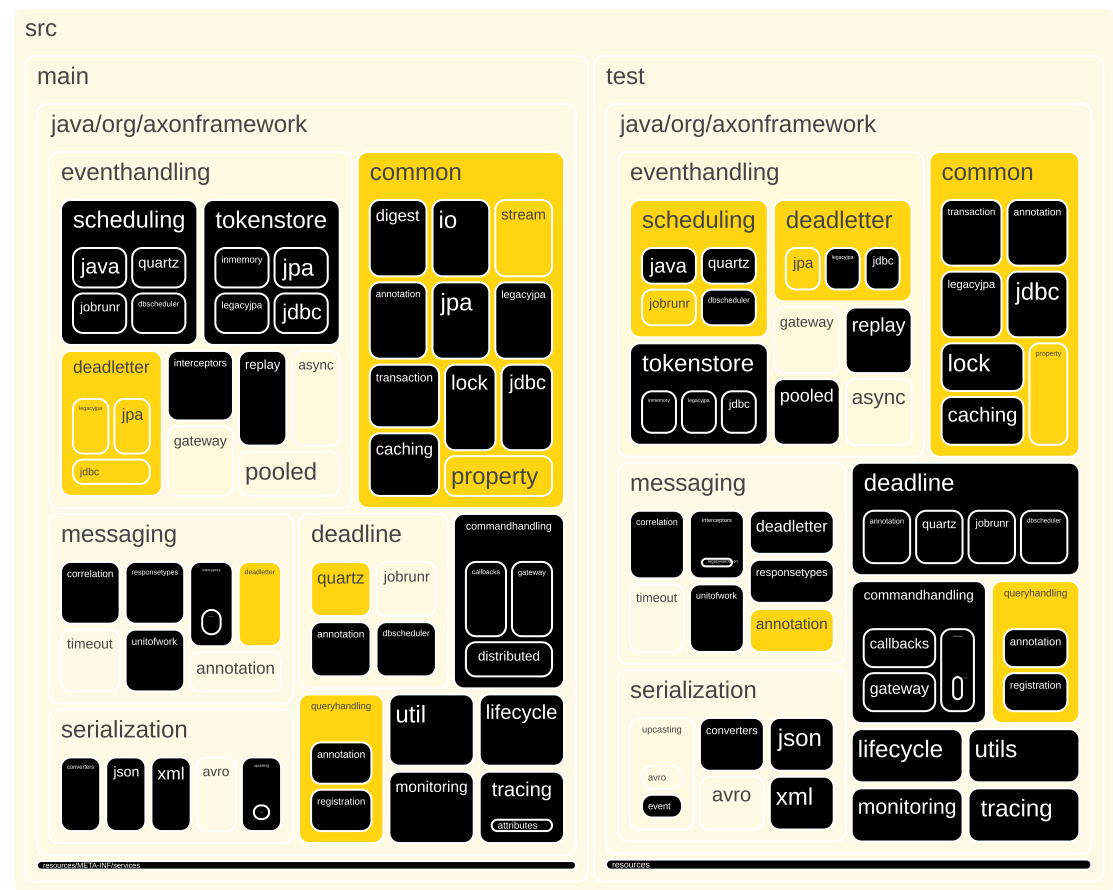
Second author with the second highest number of commits



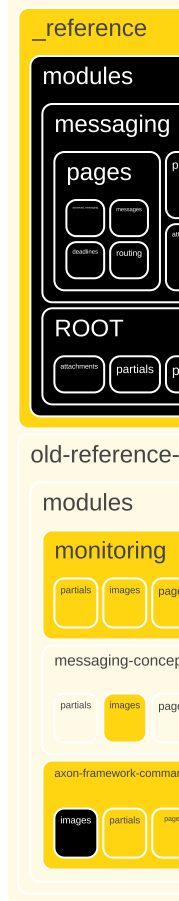
Days since last commit

AxonFramework-4.11.1

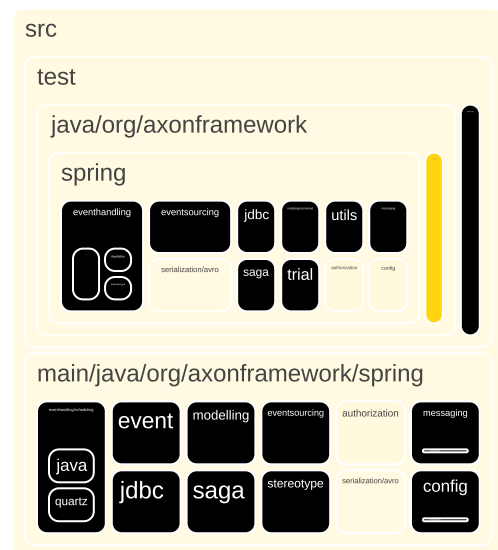
messaging



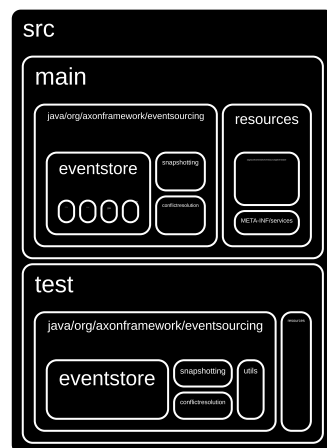
docs



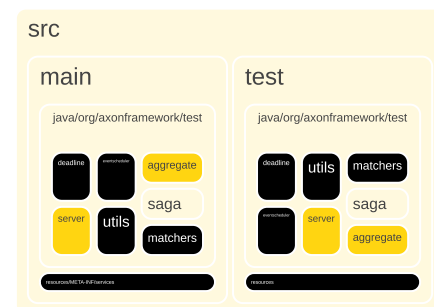
spring



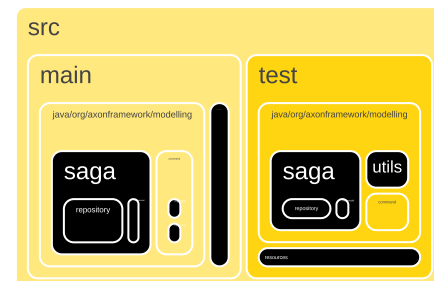
eventsourcing



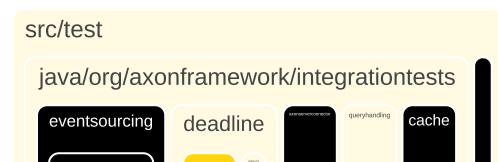
test



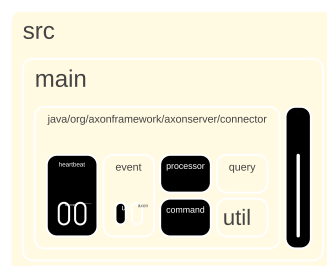
modelling



integrationtests

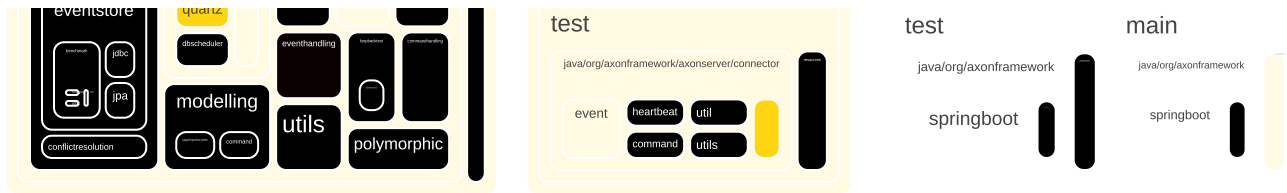


axon-server-connector



spring-boot-autoconfigure

src

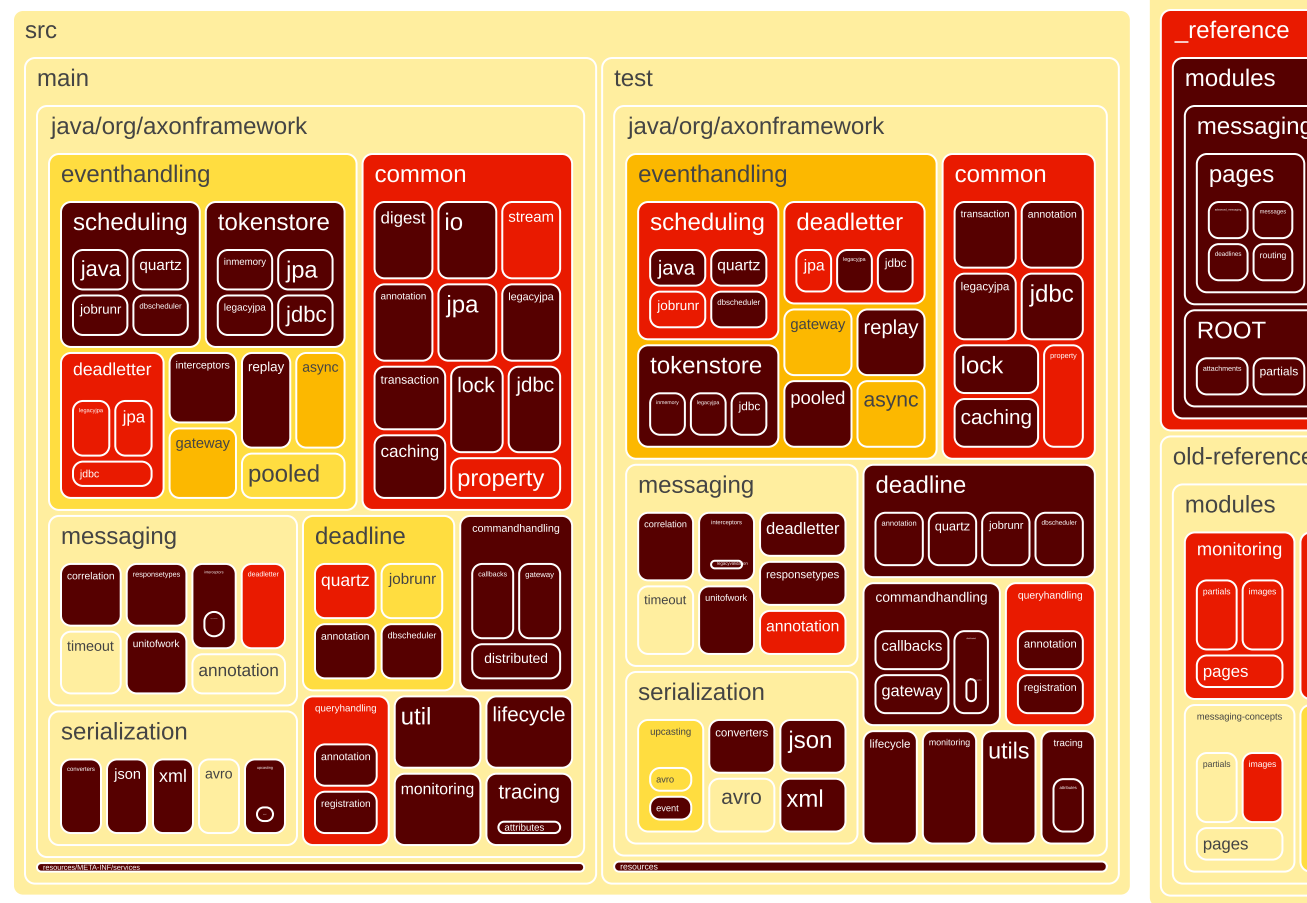


Days since last commit per directory (ranked)

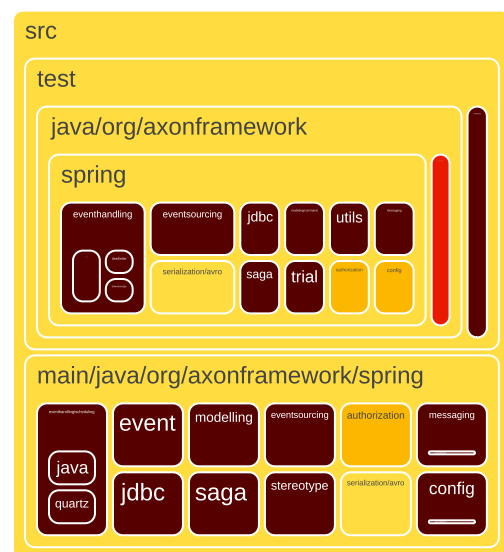
Rank of days since last commit

AxonFramework-4.11.1

messaging



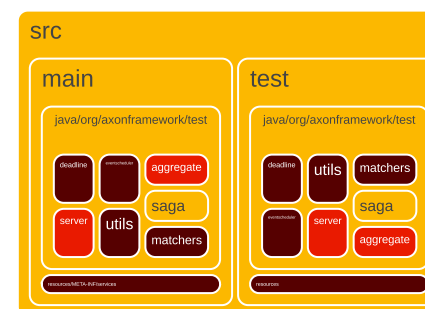
spring



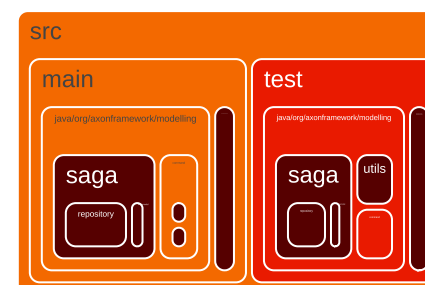
eventsourcing



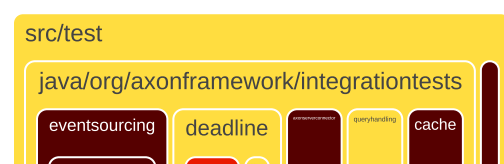
test



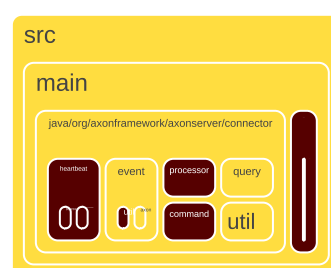
modelling



integrationtests

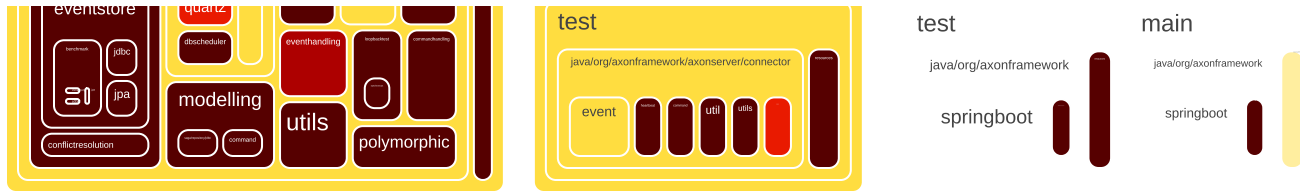


axon-server-connector



spring-boot-autoconfigure

src



Days since last file creation per directory

Days since last file creation

AxonFramework-4.11.1

messaging

src

main

java/org/axonframework

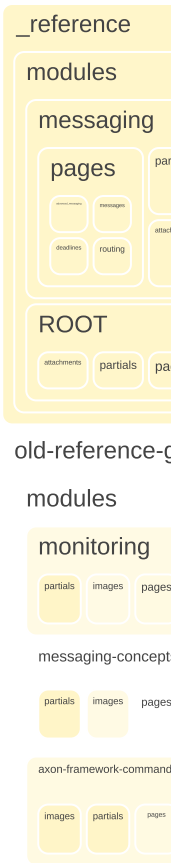


test

java/org/axonframework



docs



spring

src

test

java/org/axonframework

spring



main/java/org/axonframework/spring



integrationtests

src/test

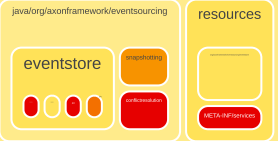
java/org/axonframework/integrationtests



eventsourcing

src

main



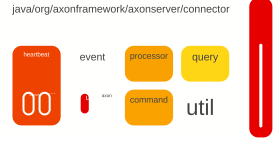
test



axon-server-connector

src

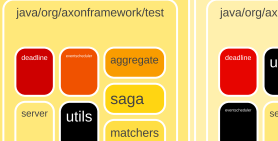
main



test

src

main



modelling

src

main



spring-boot-autoconfigure

src

Days since last file creation per directory (ranked)

Rank of days since last file creation

AxonFramework-4.11.1

messaging

src

main

java/org/axonframework

eventhandling

scheduling

java quartz

jobrunr dbcheduler

tokenstore

inmemory jpa

legacypa jdbc

common

digest io stream

annotation jpa legacypa

transaction lock jdbc

caching property

deadletter

jpa

jdbc

interceptors

replay

async

gateway

pooled

messaging

correlation

responsetypes

timeout

unitwork

annotation

deadline

quartz jobrunr

annotation dbcheduler

commandhandling

callbacks gateway

distributed

serialization

converters json xml avro

event

queryhandling

annotation

registration

util

monitoring

tracing

resources/META-INF/resources

test

java/org/axonframework

eventhandling

scheduling

java quartz

jobrunr dbcheduler

deadletter

jpa

jdbc

gateway

replay

pooled

async

tokenstore

inmemory jpa jdbc

messaging

correlation

responsetypes

timeout

unitwork

annotation

deadline

annotation quartz jobrunr dbcheduler

commandhandling

callbacks gateway

distributed

queryhandling

annotation

registration

serialization

upcasting converters json

avro

event

lifecycle

utils

monitoring

tracing

resources

spring

src

test

java/org/axonframework

spring

eventhandling

eventsourcing

serialization/avro

saga

trial

util

event

jdbc

saga

stereotype

serialization/avro

config

main/java/org/axonframework/spring

java

quartz

jdbc

saga

stereotype

serialization/avro

config

integrationtests

src/test

java/org/axonframework/integrationtests

eventsourcing

deadletter

queryhandling

cache

eventsourcing

src

main

java/org/axonframework/eventsourcing

eventstore

snapshooting

serialization/avro

saga

trial

util

event

jdbc

saga

stereotype

serialization/avro

config

axon-server-connector

src

main

java/org/axonframework/axonserver/connector

heartbeat

event

processor

query

command

util

test

src

main

java/org/axonframework/test

deadline

responsetypes

server

utils

saga

matchers

aggregate

modelling

src

main

java/org/axonframework/modelling

saga

utils

matchers

aggregate

spring-boot-autoconfigure

src

docs

_reference

modules

messaging

pages

messages

deadlines

routing

ROOT

attachments

partials

old-reference-

modules

monitoring

partials

images

pages

messaging-concept

partials

images

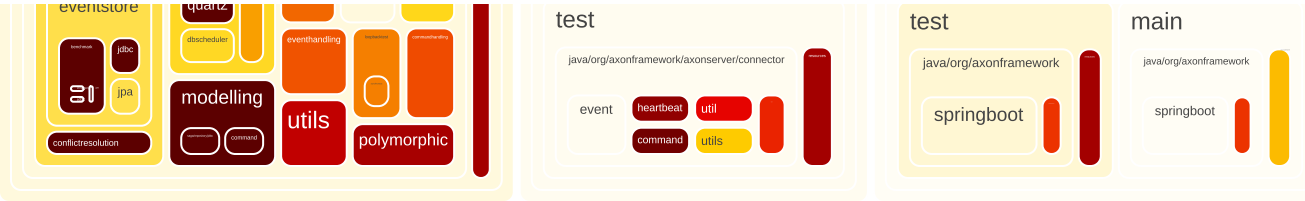
pages

axon-framework-command

images

partials

pages



Days since last file modification per directory

Days since last file modification

AxonFramework-4.11.1

messaging

src

main

java/org/axonframework



test

java/org/axonframework



docs

_reference

modules



old-reference-c

modules



spring

src

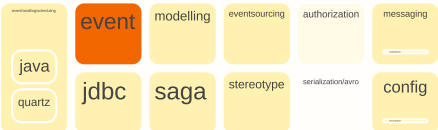
test

java/org/axonframework

spring



main/java/org/axonframework/spring



integrationtests

src/test

java/org/axonframework/integrationtests



eventsourcing



axon-server-connector

src

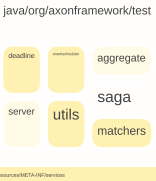
main



test

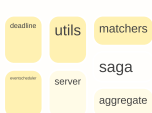
src

main



test

java/org/axonframework/test



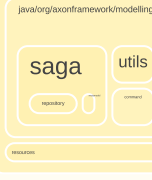
modelling

src

main

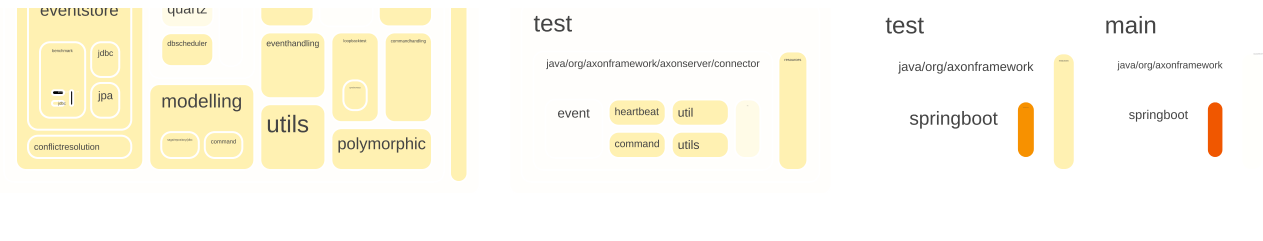


test



spring-boot-autoconfigure

src

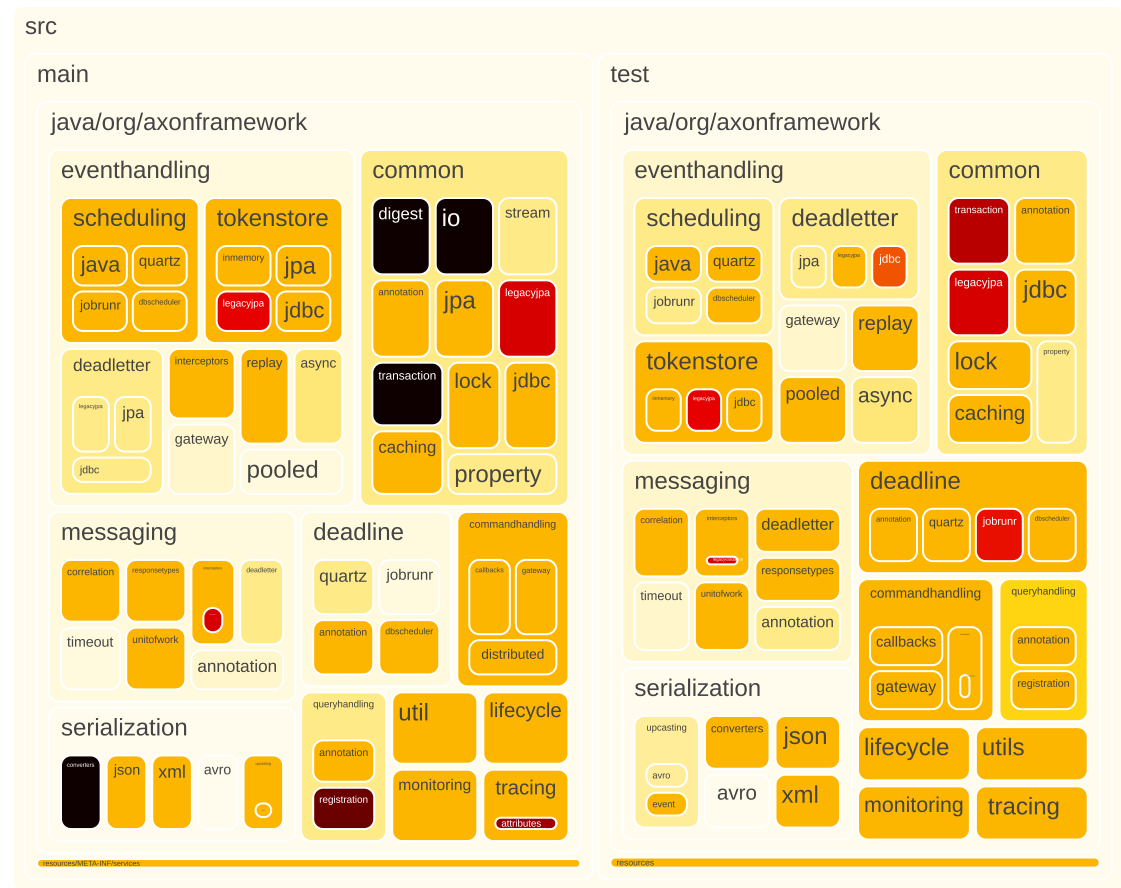


Days since last file modification per directory (ranked)

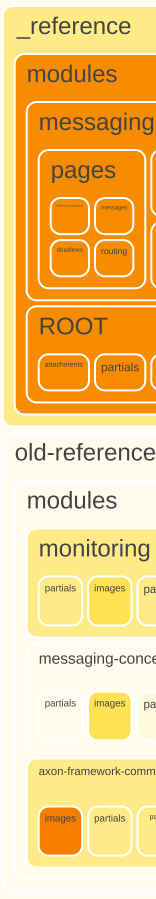
Rank of days since last file modification

AxonFramework-4.11.1

messaging



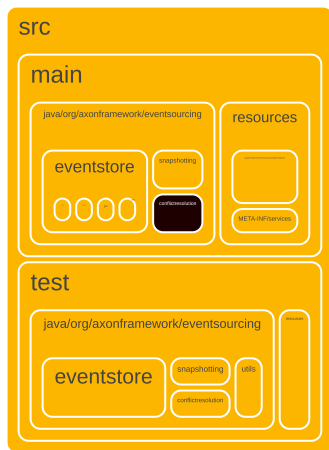
docs



spring



eventsourcing



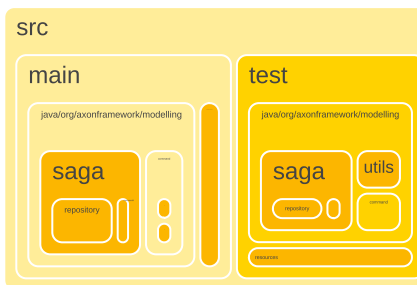
axon-server-connector



test

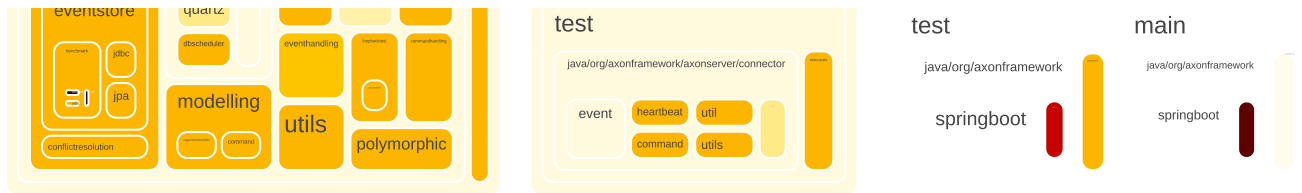


modelling



spring-boot-autoconfigure

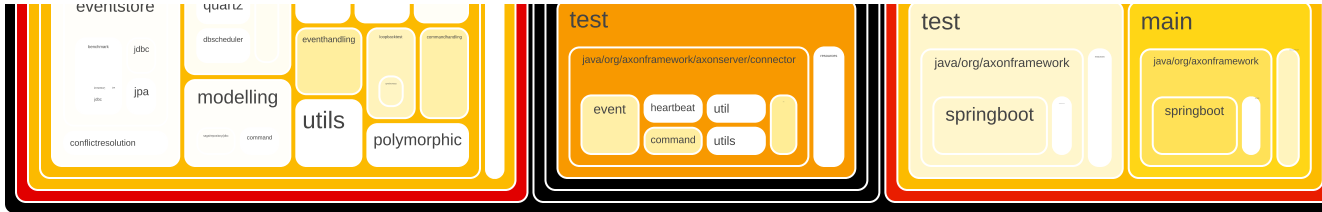
src



File changed frequently with other files

Pairwise file changes





Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

Preview data

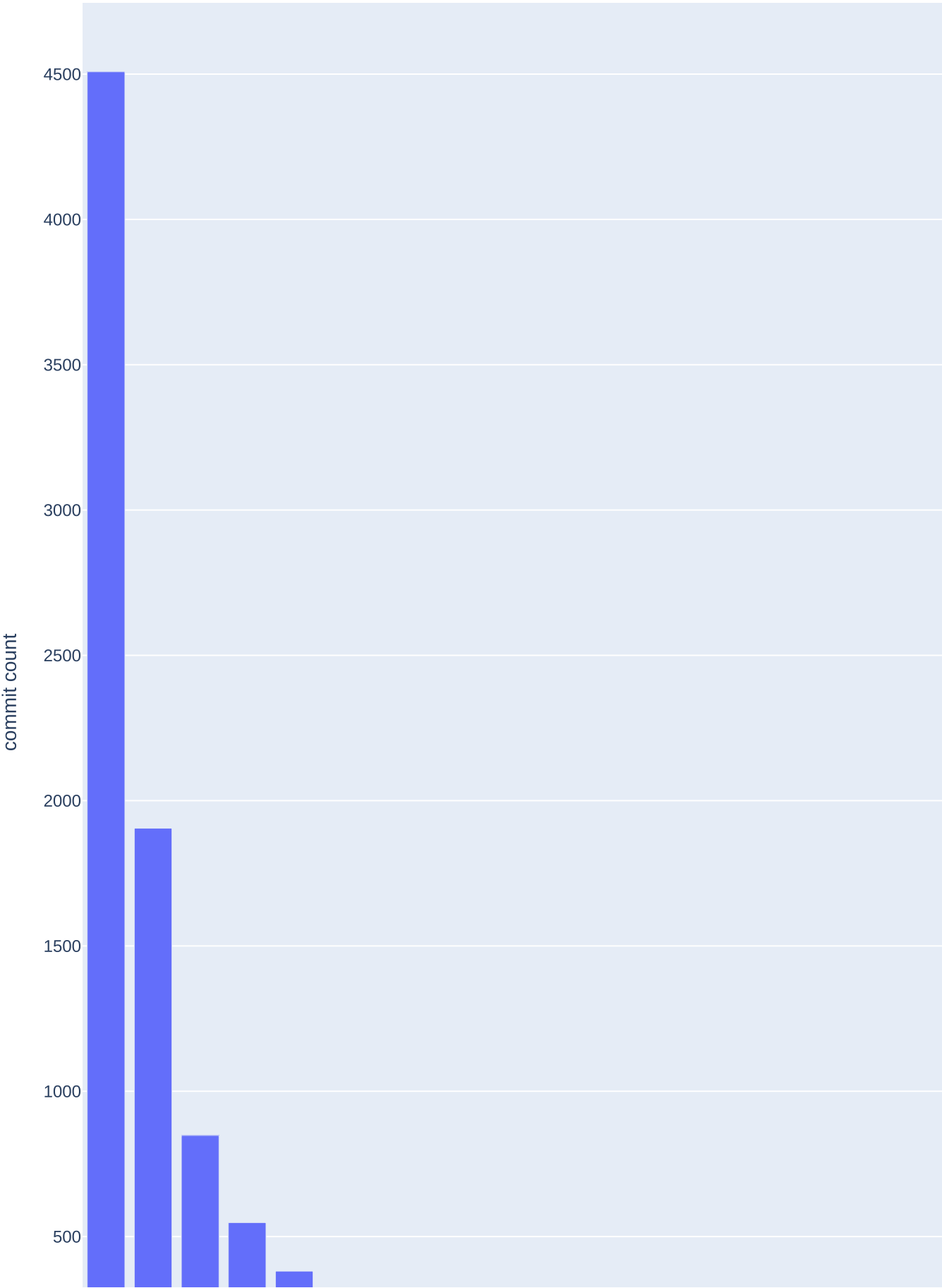
Sum of commits that changed more than 30 files (each) = 961
Max changed files with one commit = 2178

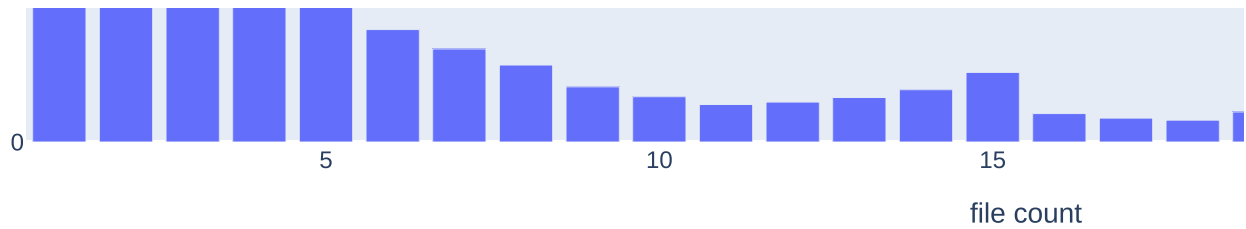
	filesPerCommit	commitCount
count	333.000000	333.000000
mean	344.075075	33.960961
std	440.375494	274.606293
min	1.000000	1.000000
25%	84.000000	1.000000
50%	186.000000	2.000000
75%	428.000000	5.000000
max	2178.000000	4508.000000

	filesPerCommit	commitCount
0	1	4508
1	2	1905
2	3	848
3	4	548
4	5	381
5	6	273
6	7	226
7	8	187
8	9	134
9	10	110
10	11	91
11	12	97
12	13	108
13	14	127
14	15	169
15	16	69
16	17	58
17	18	53
18	19	73
19	20	55
20	21	38
21	22	63
22	23	30
23	24	31
24	25	37
25	26	28
26	27	28
27	28	24
28	29	28
29	30	21

Bar chart with the number of files per commit distribution

Changed files per commit





Pairwise Changed Files vs. Dependency Weight

This section explores the correlation between how often pairs of files are changed together (common commit count) and their dependency weight. Note that these results should be interpreted cautiously, as comparing pairwise changes and dependencies is inherently challenging.

Considerations

- **Historical vs. Current State:** Pairwise changes reflect the entire git history, while dependency weight represents the current state of the codebase.
- **Commit Granularity:** Developers may use different commit strategies, such as squashing changes into a single commit or creating fine-grained commits. Ideally, each commit should represent a single semantic change for accurate analysis.
- **Dependency Representation:** Some file types (e.g., Java files with import statements) clearly define dependencies, while others (e.g., shell scripts, XML, YAML) lack explicit dependency relationships.
- **Repository Characteristics:** Repositories with generated code may have many large commits, while stabilized repositories may only update configuration files for dependency changes.

Data Preview

Received notification from DBMS server: {severity: WARNING} {code: Neo.ClientNotification.Statement.UnknownPropertyKeyWarning} {category: UNRECOGNIZED} {title: The provided property key is not in the database} {description: One of the property names in your query is not available in the database, make sure you didn't misspell it or that the label is available when you run this statement in your application (the missing property name is: cardinality)} {position: line: 9, column: 48, offset: 576} for query:
 "// List pair of files that were changed together and that have a declared dependency between each other.\n\n MATCH (firstCodeFile:File)-[dependency:DEPENDS_ON]->(secondCodeFile:File)\n MATCH (firstCodeFile)-[pairwiseChange:CHANGED_TOGETHER_WITH]-(secondCodeFile)\n //De-duplicating the pairs of files isn't necessary, because the dependency relation is directed.\n //WHERE elementId(firstCodeFile) < elementId(secondCodeFile)\n WITH firstCodeFile.fileName AS firstFileName\n ,secondCodeFile.fileName AS secondFileName\n ,coalesce(dependency.weight, dependency.cardinality) AS dependencyWeight\n ,pairwiseChange.commitCount AS commitCount\n ,dependency.fileDistanceAsFewestChangeDirectoryCommands AS fileDistanceAsFewestChangeDirectoryCommands\n RETURN dependencyWeight\n ,commitCount\n ,fileDistanceAsFewestChangeDirectoryCommands\n // ,count(*) AS occurrences\n // ,collect(firstFileName + ' -> ' + secondFileName)[0..3] AS examples\n\n ORDER BY dependencyWeight, commitCount\n\n // MATCH (firstCodeFile:File)-[dependency:DEPENDS_ON]->(secondCodeFile:File)\n // MATCH (firstCodeFile)-[pairwiseChange:CHANGED_TOGETHER_WITH]-(secondCodeFile)\n // WHERE elementId(firstCodeFile) < elementId(secondCodeFile)\n // RETURN firstCodeFile.fileName AS firstFileName\n // ,secondCodeFile.fileName AS secondFileName\n // ,dependency.weight AS dependencyWeight\n // ,pairwiseChange.commitCount AS commitCount\n // ORDER BY dependencyWeight, commitCount\n\n // MATCH (g1:!Git&File)-[relation:CHANGED_TOGETHER_WITH|DEPENDS_ON]-(g2:!Git&File)\n // WITH count(DISTINCT relation) AS relatedFilesCount\n // ,collect(DISTINCT relation) AS relations\n // UNWIND relations AS relation\n // WITH relatedFilesCount\n // ,coalesce(relation.commitCount, 0) AS commitCount\n // ,coalesce(relation.weight, 0) AS dependencyWeight\n // ,coalesce(relation.fileDistanceAsFewestChangeDirectoryCommands, 0) AS fileDistanceAsFewestChangeDirectoryCommands\n // RETURN dependencyWeight\n // ,commitCount\n // ,fileDistanceAsFewestChangeDirectoryCommands\n // ORDER BY dependencyWeight, commitCount"

	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
0	1	12	0
1	1	12	0
2	1	12	4
3	1	12	0
4	1	12	0
5	1	12	4
6	1	12	0
7	1	12	2
8	1	12	0
9	1	12	4
10	1	13	0
11	1	13	4
12	1	13	0
13	1	13	4
14	1	14	4
15	1	14	2
16	1	15	0
17	1	15	2
18	1	15	0
19	1	16	0

Data Statistics

'Pairwise changed git files compared to dependency weights - Overall statistics'

	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
count	166.000000	166.000000	166.000000
mean	9.590361	25.204819	1.373494
std	19.230680	16.573260	1.570260
min	1.000000	12.000000	0.000000
25%	1.250000	14.250000	0.000000
50%	3.000000	19.000000	0.000000
75%	10.000000	28.000000	2.000000
max	154.000000	95.000000	4.000000

'Pairwise changed git files compared to dependency weights - Pearson Correlation'

	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
dependencyWeight	1.000000	0.200368	-0.141013
commitCount	0.200368	1.000000	-0.145947
fileDistanceAsFewestChangeDirectoryCommands	-0.141013	-0.145947	1.000000

'Pairwise changed git files compared to dependency weights - Spearman Correlation'

	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
dependencyWeight	1.000000	0.214529	-0.091448
commitCount	0.214529	1.000000	-0.133180
fileDistanceAsFewestChangeDirectoryCommands	-0.091448	-0.133180	1.000000

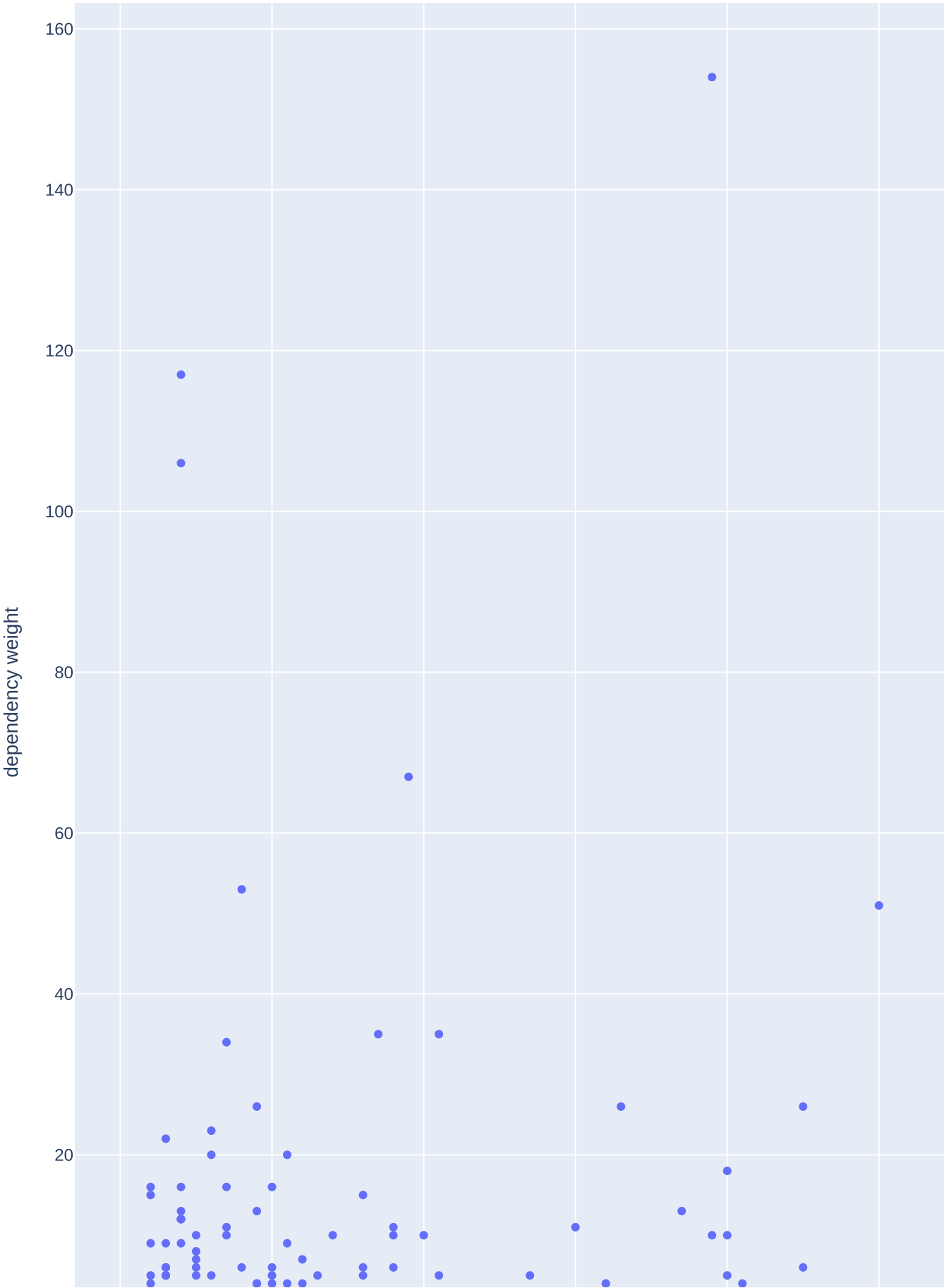
'Pearson Correlation with p-value for commitCount and dependencyWeight'

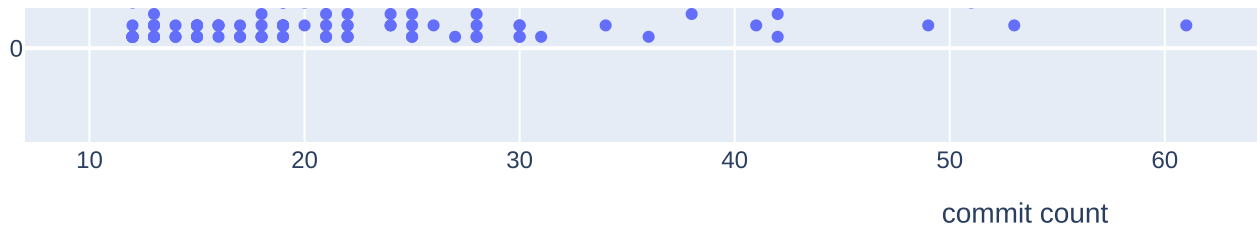
PearsonRResult(statistic=0.20036758755264605, pvalue=0.009643647845018762)

'Spearman Correlation with p-value for commitCount and dependencyWeight'

SignificanceResult(statistic=0.21452881933918705, pvalue=0.005511439972198768)

Pairwise changed files: Number of changes (commitCount) vs. dependency weight





WordCloud of git authors

word frequency		
0	Steven van Beelen	3360
1	Allard Buijze	2721
2	smcvb	822
3	Milan Savic	449
4	Mitchell Herrijgers	309
5	Gerard Klijs	246
6	Marc Gathier	231
7	sara pellegrini	163
8	Rene de Waele	121
9	Jетро Coenradie	117

Wordcloud of git authors

